

Zimeng Wang

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Education

- Aalto University**, M.Sc. in Control, Robotics and Autonomous Systems Sept 2024 – Present
- GPA: 4.83/5.0
 - **Honor: Full Tuition Fee Scholarship**; 2nd Award LeRobot Hackathon(**Top 5** Worldwide)
 - **Thesis:** Self-supervised Configuration Space Distance Function
- École polytechnique fédérale de Lausanne(EPFL)**, Exchange Student Sept 2025 – Dec 2026
- GPA: 5.22/6.0
 - **Honor:** Mobility Scholarship
- Southern University of Science and Technology**, B.Eng in Robotics Engineering Sept 2020 – Jun 2024
- GPA: 3.73/4.0 (**Top 10%**)
 - **Honor:** 3rd Award of Merit Student Scholarship (2022,2023); 2nd Prize of MEICC (2022, National 10–25%)
 - **Thesis:** Research on Control Algorithms for the Wheel-Legged Robot Achilles-1 (*under Prof. Kemi Ding*)

Research Experience

- A Comparative Analysis of Optimization-Based and Reactive Dynamical Systems** Sept 2025 – Present
- Visiting Student Research Project, LASA Lab, EPFL
Supervisors: *Prof. Aude Billard* and *Dr. Rui Wu*
- Established a comprehensive benchmark comparing GPU-accelerated MPPI control (NVIDIA CuRobo) against geometrically-guided hybrid Dynamical Systems for reactive motion planning.
 - Developed a hybrid motion generation framework for a 7-DoF Franka Emika Panda, integrating **Diffeomorphic Transformations** and implicit representations (**Neural-JSDF/CDF**) to guarantee stable obstacle avoidance.
- Self-supervised Configuration Space Distance Function** Feb 2026 – Dec 2026
- Master's Thesis Student, Robot Learning & Interaction Group, Idiap/EPFL
Supervisors: *Dr. Sylvain Calinon*, *Prof. Dominik Baumann* and *Yiming Li*
- Developed SCDF, a self supervised physically informed neural network that removes the expensive C-space ground-truth labeling required by the original CDF while preserving **one step IK** solution.
 - Integrated SCDF into learning based and gradient based pipelines, enabling controlled comparisons against neural distance functions(**JSDF, CDF, RDF, no-distance**), and **PointNet** baselines.
- Scalability of Multi-Agent Path Finding** Sept 2024 – Jun 2025
- Research Assistant, Mobile Robotics Group, Aalto University
Supervisor: *Prof. Tomasz Kucner*
- Introduced and optimized Hybrid Velocity Obstacle (HVO) methods, enhancing multi-agrnt obstacle avoidance efficiency and local decision-making.
 - Conducted simulations with 12 agents, achieving a **10% improvement** in overall planning efficiency compared to global planners (measured by total agent planning time).

Projects

- RoboMaster Robotics Competition [Video]** Sept 2023 – Jul 2024
- Simulation and Control Engineer, SUSTech ARTINX Robotics Lab
- Developed a wheel-legged robot capable of smooth ground movement and obstacle traversal for competition; responsible for its simulation and control algorithms.
 - Utilized the dynamics to build the state-space equation of the robot; used **Linear Quadratic Regulator (LQR)** and **Virtual Model Control (VMC)**for optimal control of dynamic systems.

- Competed in the DJI RoboMaster Robotics Competition, advanced to the national **Top 32**, and further developed the work into the undergraduate thesis project.

Design of a Lobster-like Lychee-Picking Robot

Jul 2022 – Aug 2022

Individual Researcher *under Prof. Chaoyang Song*

- Designed a lychee-picking robot imitating lobster joints and Bennett joints to realize functions of cutting, clipping, and pulling in one tool.
- Conducted the inverse kinematics analysis and generative design for the manipulator.
- Participated in the MEICC and achieved **National 2nd Prize**.

Additional Information

Languages: Mandarin (native); English (IELTs 7.5)

Programming Language: Python, Java, C/C++

Programming Framework: ROS, PyTorch, cuRobo, IsaacLab

Software: Matlab, Solidworks, Fusion360, Keil